state Vector:

$|0\rangle$ ________

$$|\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$
$$= \begin{bmatrix} a \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ b \end{bmatrix}$$
$$= a\begin{bmatrix} 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} 0 \\ 1 \end{bmatrix}$$
$$= a|0\rangle + b|1\rangle$$

$|a|^2 + |b|^2 = 1$

$p(0)$ $p(1)$

Unitary matrix:

$$U_{n \times n} = \begin{bmatrix} & & \end{bmatrix}$$

$U^\dagger = U^{-1}$

$U^\dagger U = I$ $\quad$ $UU^\dagger = I$

Single qubit gates:

logical gates:

NOT gate

I	O
0	1
1	0

$0 \xrightarrow{NOT} 1 \xrightarrow{NOT} 0$ (Reversable)

Reset zero

I	O
0	0
1	0

$0 \xrightarrow{R} 0 \xrightarrow{R} 0$

$1 \xrightarrow{R} 0 \xrightarrow{R} 0$ (Not Reversable)

In quantum gates it should be reversable "each input should have a distinct output".

pauli gates:

$$\underset{\sigma_x}{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$\underset{\sigma_y}{Y} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$\underset{\sigma_z}{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$X|0\rangle = |1\rangle$$

* Matrix Mul $\quad \underset{2\times 2}{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}} \underset{2\times 1}{\begin{bmatrix} 1 \\ 0 \end{bmatrix}} = \underset{2\times 1}{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}$

$$Z|0\rangle = |0\rangle \qquad \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$Z|1\rangle = -|1\rangle$$

σ is Unitary: $\sigma^\dagger \sigma = I \qquad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

σ is Hermition: $\sigma^\dagger = \sigma$

$$X\,X|0\rangle = X|1\rangle = |0\rangle$$

$$|\psi\rangle = a|0\rangle + b|1\rangle = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$X|\psi\rangle = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} b \\ a \end{bmatrix}$$

$$|\psi'\rangle = b|0\rangle + a|1\rangle$$

$$X|\psi\rangle = aX|0\rangle + bX|1\rangle$$
$$= a|1\rangle + b|0\rangle$$

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

$$= |1\rangle\langle 0| + |0\rangle\langle 1|$$

outer product

$$\begin{bmatrix} 0 \\ 1 \end{bmatrix}\begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$X|0\rangle = \left(|1\rangle\langle 0| + |0\rangle\langle 1|\right)|0\rangle$$
$$= |1\rangle\langle 0|0\rangle + |0\rangle\langle 1|0\rangle$$
$$= |1\rangle + 0 = \underline{|1\rangle}$$

outer product:

output $\qquad\qquad$ input

$$|\ \rangle\langle\ |$$

$$Z = |0\rangle\langle 0| - |1\rangle\langle 1|$$

$$= \begin{bmatrix} 1 \\ 0 \end{bmatrix}\begin{bmatrix} 1 & 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix}\begin{bmatrix} 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} - \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = Z$$

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$H|0\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right) = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$$

$$= |+\rangle$$

$$H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$$

$$H = |+\rangle\langle 0| + |-\rangle\langle 1|$$

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

$$p(0) = \frac{1}{2} \qquad p(1) = \frac{1}{2}$$

superposition

$$HH|\psi\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix}$$

$$= \frac{1}{2}\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix} = |\psi\rangle$$

Rotation gates:

Rotate by angle θ on pauli axis

$R_x(\theta) \qquad R_y(\theta) \qquad R_z(\theta)$

$$I\cos\left(\frac{\theta}{2}\right) - i\,\sigma\sin\left(\frac{\theta}{2}\right)$$

For $\sigma_x \Rightarrow R_x$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\cos\left(\frac{\theta}{2}\right) - i\sin\left(\frac{\theta}{2}\right)\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -i\sin\left(\frac{\theta}{2}\right) \\ -i\sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{bmatrix}$$

For $\theta = \pi$

$$R_x(\pi) = \begin{bmatrix} \cos\left(\frac{\pi}{2}\right) & -i\sin\left(\frac{\pi}{2}\right) \\ -i\sin\left(\frac{\pi}{2}\right) & \cos\left(\frac{\pi}{2}\right) \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -i \\ -i & 0 \end{bmatrix} = -i\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

For σ_y

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\cos\left(\frac{\theta}{2}\right) - i\sin\left(\frac{\theta}{2}\right)\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{bmatrix}$$

For $\theta = \pi$ $\quad R_y(\pi) = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

For σ_z

$$R_z = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \cos\left(\frac{\theta}{2}\right) - i\sin\left(\frac{\theta}{2}\right) \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} \cos\left(\frac{\theta}{2}\right) - i\sin\left(\frac{\theta}{2}\right) & 0 \\ 0 & \cos\left(\frac{\theta}{2}\right) + i\sin\left(\frac{\theta}{2}\right) \end{bmatrix}$$

$$\cos(x) - i\sin(x) = e^{-ix}$$

$$\cos(x) + i\sin(x) = e^{ix}$$

Euler Formula

$$R_z(\theta) = \begin{bmatrix} e^{-i\frac{\theta}{2}} & 0 \\ 0 & e^{i\frac{\theta}{2}} \end{bmatrix}$$

For $\theta = \pi$

$$R_z(\pi) = \begin{bmatrix} e^{-i\frac{\pi}{2}} & 0 \\ 0 & e^{i\frac{\pi}{2}} \end{bmatrix}$$

$$= \begin{bmatrix} -i & 0 \\ 0 & i \end{bmatrix}$$

Any gate can be created by Rotation gates

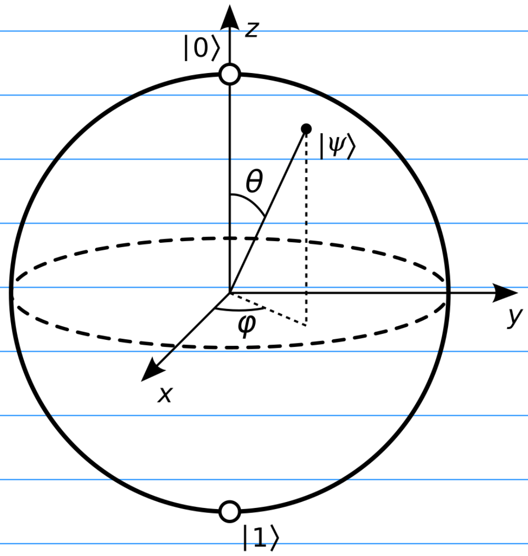
$$U = e^{i\alpha} R_z(\beta) R_y(\gamma) R_z(\delta)$$

For $\alpha = \frac{\pi}{2}$ $\beta = 0$ $\lambda = \frac{\pi}{2}$ $\delta = \pi$

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Bloch sphere:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$



$r_x = \sin(\theta)\cos(\phi)$

$r_y = \sin(\theta)\sin(\phi)$

$r_z = \cos(\theta)$

(r_x, r_y, r_z) $\qquad$ $r_x^2 + r_y^2 + r_z^2$

$$\sin(\theta)^2\cos(\phi)^2 + \sin(\theta)^2\sin(\phi)^2 + \cos(\theta)^2$$

$$= \sin(\theta)^2\left(\cos(\phi)^2 + \sin(\phi)^2\right) + \cos(\theta)^2$$

$$= \sin(\theta)^2 + \cos(\theta)^2 = 1$$

$(1,0,0)$ $\quad r_x = 1 \quad r_y = 0 \quad r_z = 0$

$\theta = \frac{\pi}{2} \quad \phi = 0$

$$|\psi\rangle = \cos\left(\frac{\pi}{4}\right)|0\rangle + e^{i\cdot 0}\sin\left(\frac{\pi}{4}\right)|1\rangle$$

$$= \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle$$ X axis pos

$|-\rangle$ X axis Neg

For Z axis $|0\rangle, |1\rangle$

For X axis $|+\rangle, |-\rangle$

For y axis $|+i\rangle, |-i\rangle$

$$\frac{|0\rangle + i|1\rangle}{\sqrt{2}}, \quad \frac{|0\rangle - i|1\rangle}{\sqrt{2}}$$

Applying single gates:

$|0\rangle$ —$\boxed{X}$— $|\psi\rangle$

$$X|0\rangle = |1\rangle$$

$|0\rangle$ — $\boxed{H}$ — $\boxed{Z}$ — $|\psi\rangle$

$$H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}} = |+\rangle$$

$$= |\psi_0\rangle$$

$$Z|\psi_0\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$$

$$ZH|0\rangle =$$

$$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}|0\rangle =$$

$$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}|0\rangle =$$

$$\frac{|0\rangle - |1\rangle}{\sqrt{2}} = |-\rangle$$

$$HXH|0\rangle$$

$H|\psi\rangle =$